Start Here

SUBStart Here

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Meeting folder

Everything needed to discuss this work, in one place. Self-contained — nothing here depends on the rest of the repository, though it points at it.

Read in this order

Everything is a PDF. The `.md` files beside them are the editable sources; rebuild all PDFs with `python make_pdfs.py`.

#	file	what it is	time
0	`00-plain-english.pdf`	The whole thing with no jargon, and a glossary	6 min
1	`02-findings.pdf`	The findings, technical	10 min
2	`03-the-ask.pdf`	The one decision I need	2 min
3	`papers/01-mittelstadt-2023-*.pdf`, section 6 only	The closest prior work	15 min
4	`04-paper-draft.pdf`	The full paper draft	40 min
5	`05-reading-notes.pdf`	What each cited paper was checked for	10 min

`01-start-here.pdf` is this document. **Start at 00 if any of the terminology is unfamiliar** — it defines everything the other documents assume.

The one-sentence version

> Levelling down is not what fairness constraints do — it is what they do below a selection rate of about 0.3. Above it, the same constraint hands decisions out instead of taking them away. And nearly every real deployment sits below it.

The papers, and why each is here

file	why it matters to this work
01 Mittelstadt et al. (2023)	**The closest prior work.** Established levelling down *and* proposed the remedy (§6, "minimum rate constraints"). Read §6.
02 Corbett-Davies et al. (2017)	The optimal fair classifier is group-specific thresholds — makes our comparison arm a *bound*, not a rival.
03 Diana et al. (2021)	Minimax group fairness: the established levelling-down remedy we benchmark against.
04 Kearns et al. (2018)	Fairness gerrymandering — our subgroup finding is an empirical replication.
05 Goethals et al. (2024)	**The nearest miss.** Studies the selection-rate axis, calls it "overlooked" — and two of its four authors also wrote 01. They fix the pool as a budget, so direction cannot move.
06 Menon & Williamson (2018)	Thresholding characterisation; supporting cite for 02.
07 Ustun et al. (2019)	Decoupled classifiers with preference guarantees.
08 Black et al. (2022)	Model multiplicity — supports our arbitrariness caution.
09 Agarwal et al. (2018)	The base paper. Everything is built on this reduction.
10 Ding et al. (2021)	Retiring Adult — the ACS datasets, and the argument for not trusting Adult alone.

Where the work lives

- `../docs/11–26` — 16 research documents, one per finding, each with its limits stated
- `../../docs/01–10` — the course-side work (Adult only)
- `../results/` — every number, as CSV
- `../../tests/test_documented_claims.py` — re-derives every headline figure and fails if a document and its data disagree
- `git log --oneline 28bc8d1..HEAD` — the full record, including pre-registrations committed before their experiments ran

Two things to raise before they are asked

1. The remedy is not novel. I originally believed the selection-rate floor was new. It is a variant of Mittelstadt et al.'s minimum rate constraints (paper 01, §6). I found this myself by reading the paper, corrected the write-up in place, and demoted it to a supporting result. What survives: ours is in-processing rather than post-processing, so it needs no protected attribute at prediction time; it stacks with parity rather than replacing it; and it is evaluated on held-out data across 26 populations where they report Adult's training set.

2. The mechanism failed. I derived a candidate explanation for the crossover, pre-registered it with numerical thresholds, and tested it on fourteen populations run afterwards. It cleared its bars and was beaten by a trivial constant rule. It is reported as a failure. I can say *when* the direction flips, not *why*.

Scale, for reference

26 populations · 61 experimental arms · 3 source datasets · 2 domains · 2 protected attributes · 5 seeds throughout